

# SYS-I0-AUX

Rev E0 | 2026-10-04 | Schematics authored by Chinmoy Bhuyan | System I/O and auxiliary board, 25-110 kW DC/DC family (ECO-03/04/06/07)

## 01 24 V entry, protection and window

2 cabinet feeds (LM74700 ORing) + AUX-HV feed (LM74800 switch, cabinet priority, OV), 2 mF surge reservoir, precision OV cut-off 26.13-26.42 V, PG\_24V window, eFuse UVLO

File: sheets/01\_power\_in.kicad\_sch

## 02 24 V branches and local rails

+24V to CTRL-C2000 (2 A eFuse), +24V\_GD gate-drive bias (5 A eFuse, channel B), 5 V buck (LMR36015), 3.3 V LDO, rail LEDs and test points

File: sheets/02\_rails.kicad\_sch

## 03 CTRL-C2000 and BMU-GW connectors

J301 = interfaces.SYS 2x40 (every contract signal), J302 = interfaces.GW 2x5; CANA and SCIC go to the BMU-GW socket; SCIC receiver always enabled

File: sheets/03\_ctrl\_if.kicad\_sch

## 04 Isolated CAN ports (CANB, MCAN)

CANB module bus and MCAN CAN-FD service port; ISO1042 + SN6505B isolated 5 V each, jumper-selectable split termination, ESD2CAN24 TVS, 4-pole field terminal

File: sheets/04\_can.kicad\_sch

## 05 Isolated RS-485 ports (SCIA, SCIB)

Two half-duplex RS-485 ports; ISO1410 + SN6505B isolated 5 V each, jumper-selectable 120R termination, SM712 TVS, 4-pole field terminal

File: sheets/05\_rs485.kicad\_sch

## 06 Ethernet magnetics and chassis bond

RJ45 with integrated 1:1 magnetics for the 100BASE-TX PHY on CTRL-C2000, centre taps on ETH\_CT (PHY analog supply), link / activity LEDs, PE bonding point

File: sheets/06\_eth.kicad\_sch

## 07 Isolated 24 V digital inputs

8 x IEC 61131-2 type 1/3 inputs, 4 x ISO1212; DI1 ESTOP\_A (U701), DI8 ESTOP\_B (U704) feed the chain; FB\_A/B\_MAIN (PELV GND) on J702 and in their own package U703

File: sheets/07\_di.kicad\_sch

## 08 Dual-channel safety chain and watchdog

A: E-stop A, 24 V UV, spare loop, watchdog -> latch A -> GATE\_EN, 24V\_DO switch A  
B: E-stop B, watchdog -> latch B -> CHB\_OK -> +24V\_GD, 24V\_DO switch B

File: sheets/08\_chain.kicad\_sch

## 09 Gated 24 V outputs (contactors, relays)

DO1..DO8 per interfaces.DO on 4 x TPS272C45D behind two series supply switches; switch A (channel A, DO\_PWR\_EN) and switch B (channel B, CHB\_OK); TVS coil clamps

File: sheets/09\_do.kicad\_sch

## 10 Fan channels

4 x TPS16630 fan supply (3.6 A limit, auto-retry, OV cut-off), open-drain PWM, clamped tach; sized for PFC1224DE-F00 (2.0 A typ, 2.4 A max, 26.5 V max)

File: sheets/10\_fans.kicad\_sch

## 11 NTC inputs, ADC and supply monitors

ADS7953 16-ch SAR on SPL\_CS\_ADC\_N, REF3025 2.5 V; CH0-11 = NTC1-12, CH12-14 = feed voltages, CH15 = +24V\_GD gate-drive bias current (eFuse IMON)

File: sheets/11\_ntc.kicad\_sch

## 12 Service storage and diagnostics

I2C: FM24CL64B F-RAM 0x50 (serial, calibration, fault log, hours), TMP1075 0x48; SPL\_CS\_IO\_N: two MCP23S17 (HAEN, A0 0/1): faults, rail sense, channel self-test

File: sheets/12\_service.kicad\_sch

24 V: two cabinet feeds (fused 30 A, TVS 33 V, LM74700 ideal-diode ORing) and the AUX-HV bootstrap feed (fused, TVS, LM74800 load switch with OV cut-off and a CdVdT ramp; it connects only when no cabinet feed is healthy) onto 24V\_OR with a 2 mF surge reservoir. Every outgoing branch is a TPS16630 eFuse with the shared precision OV cut-off (26.13-26.42 V, D-013) and UVLO (16 V): +24V to CTRL (2 A), +24V\_GD gate-drive bias (5 A, channel B), 24V\_DO via switch A and switch B in series (6 A), four fans (3.6 A). While AUX-HV is the source, hardware forces the fans, +24V\_GD and 24V\_DO off and latches both chain channels (D-022: AUX-HV <= 30 W; checked every build). PG\_24V = inside the 21.6-26.0 V window.

Safety chain (ECO-04), two channels on separate parts: A = E-stop A, 24 V UV, spare loop, watchdog -> latch A -> GATE\_EN + 24V\_DO switch A; B = E-stop B, watchdog -> latch B -> CHB\_OK -> +24V\_GD + 24V\_DO switch B. Each channel alone stops the gates and opens 24V\_DO; the build simulates every single stuck fault (outputs/SYS-I0-AUX\_safety\_fmea.csv).

Isolation: CANB, MCAN, RS-485 A/B each have their own SN6505B supply and ground; the DI field side (DIF\_GND) is a separate domain; PE is reached only through the RJ45 Bob-Smith node and the 470 pF DI field-ground capacitor. Budget: 16 A steady worst case, 20 A for 100 ms.

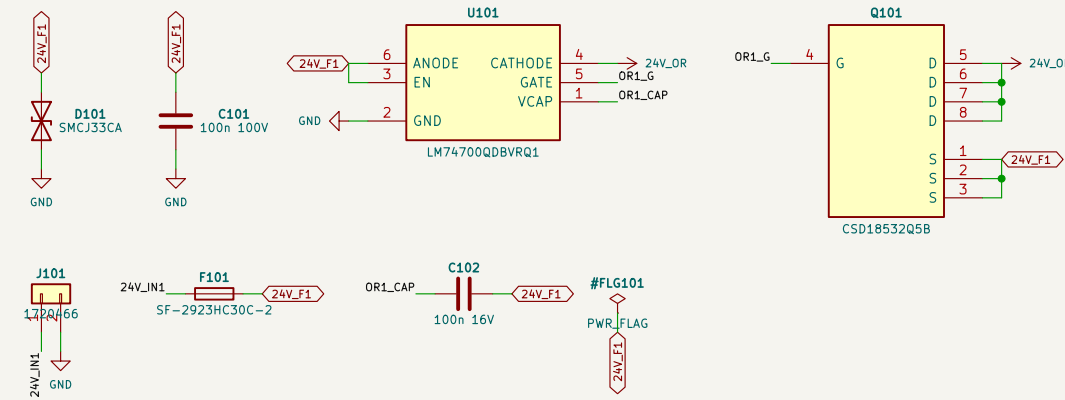
Not bench-validated. Dual-channel chain gate- and single-fault-checked on every build.  
Sheet 1  
Schematics authored by Chinmoy Bhuyan  
Domains: PELV, CANB, MCAN, RSA, RSB, DI field, PE - only isolators cross (checked)

Sheet: /  
File: SYS-I0-AUX.kicad\_sch

Title: SYS-I0-AUX - Cover / Hierarchy

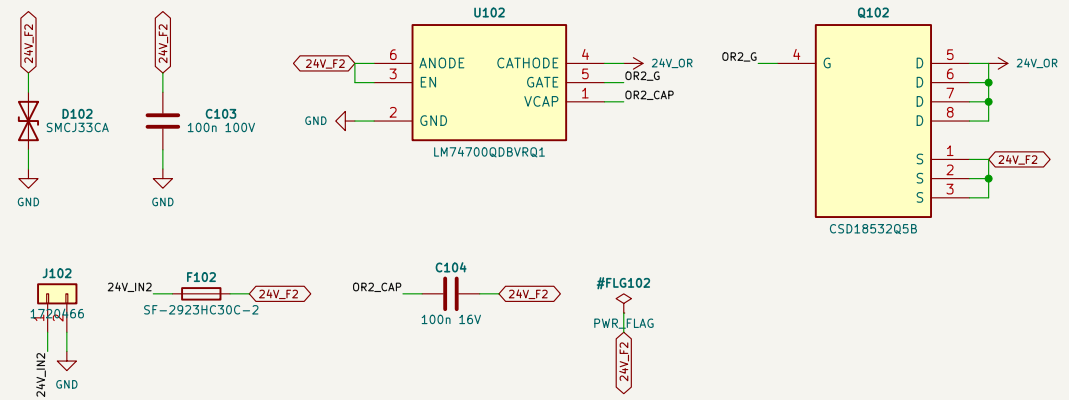
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| Size: A3            | Date: 2026-10-04 | Rev: E0  |
| KiCad E.D.A. 10.0.6 |                  | Id: 1/13 |

FEED 1: CABINET 24 V A



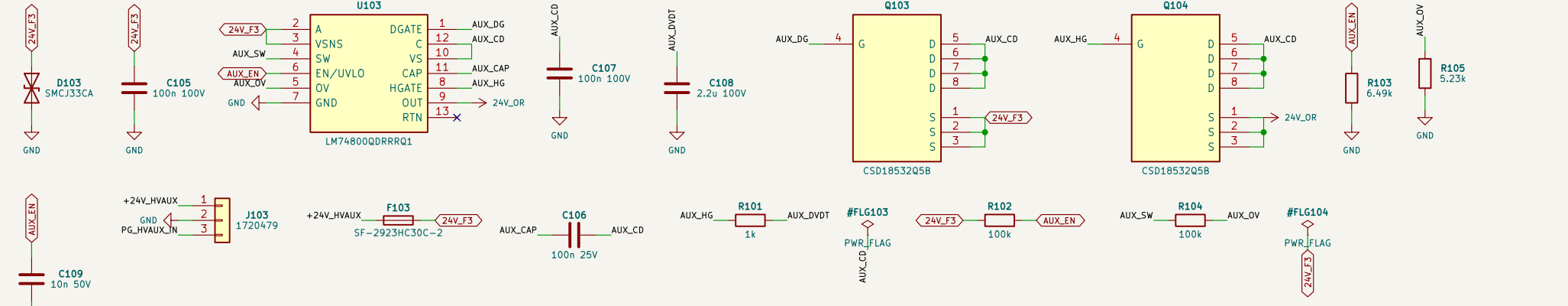
Fuse 30 A / 60 VDC: TVS 1500 W bidirectional (53 V clamp, blocks -33 V wiring error).  
LM74700 + CSD18532Q5B: ideal diode, ORs the feeds, blocks reverse current and polarity.

FEED 2: CABINET 24 V B



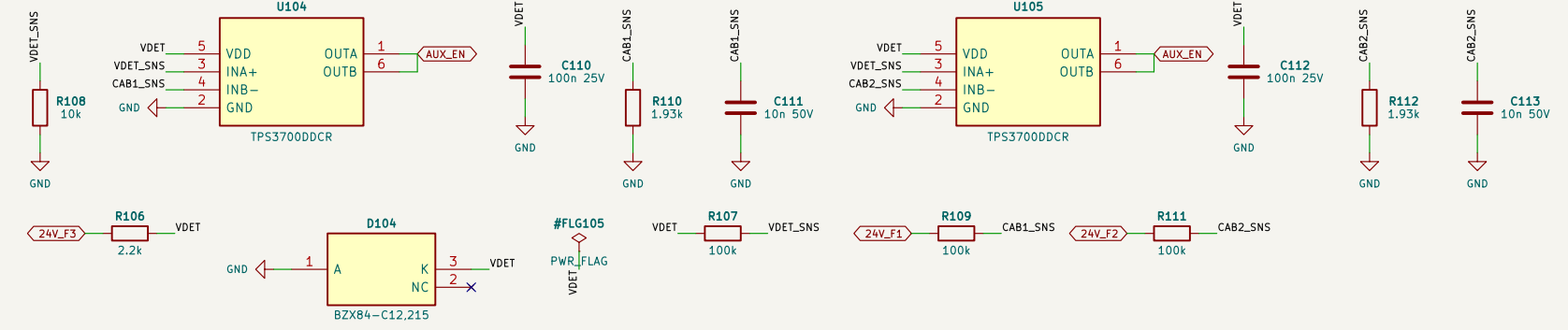
Fuse 30 A / 60 VDC; TVS 1500 W bidirectional (53 V clamp, blocks -33 V wiring error).  
LM74700 + CSD18532Q5B: Ideal diode, ORs the feeds, blocks reverse current and polarity.

FEED 3: AUX-HV 24 V (30 W BOOTSTRAP, INTERFACES.AUX)



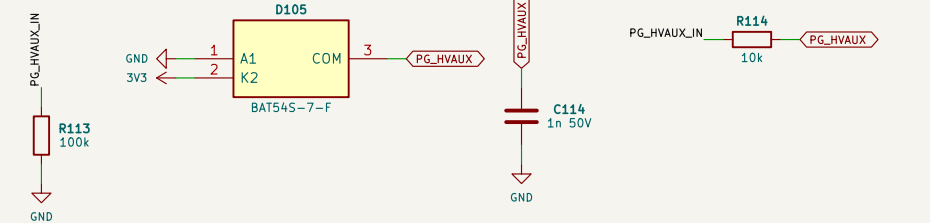
LM74800 + 2 x CSD18532Q5B back-to-back (common drain); ideal diode (DGATE) + load disconnect (HGATE).  
 Connects only when NO cabinet feed is healthy (priority detector holds EN/UVLO in shutdown); cabinet healthy  
 above 20.85 - 21.41 V rising, lost below 20.37 - 21.20 V falling (only in window: cabinet wins, INT - I/O). AUX UVLO  
 20.9 - 21.49 V rising / 18.54 - 19.46 V falling; AUX OVS 25.97 - 25.57 V, re-arm below 21.98 - 23.39 V; failed AUX-HV  
 (23.8 - 28.5 V) is cut off before the 26.13 V bus cut-off and stays off (INT-OV). AUX-HV overshoot must stay <  
 23.97 V, while AUX-EN is in, sheet 0B forces fans + 24V\_DO and 24V\_DO off, latched (D-022, < 30 W).  
 HGATE ramp (LM74800 + 10 nF to 5V), R5 at 1 Collet 220 Ohm VDR; a disabled HGATE sits at OUT, so Collet tracks the  
 bus. Black start: 0.016 - 0.063 V/ms, one ramp in 0.39 - 1.53 s, AUX-HV < 36.0 V (limit 39.8 V), < 5 W for < 60  
 ms (liner > 88 ms), Cabinet -> AUX: Inhibit < 0.17 ms, HGATE on < 1.08 ms, FEI Conducts < 5.98 ms after  
 24V\_DO is established; bus -> INT 2 V > logic UVLO 15.48 V (+24V fuse) at 27.8 V; AUX -> cabinet: AUX-HV  
 -35 uV, bus fails to the cabinet - no gap; +24V\_DO / 24V\_DO stay until FLICLR.

**CABINET PRIORITY DETECTOR (POWERED FROM THE AUX FEED)**



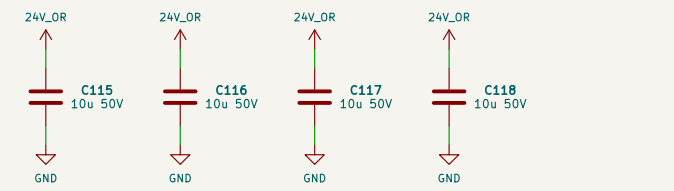
BZX84-C12 rail from 24V\_F3; works before any logic rail exists, so AUX-HV can start the module alone. Each TPS3700 OUTB pulls AUX\_EN low (LM74800 shutdown, both FETs off) while its cabinet feed is healthy. OUTA (INA+ = VDET / 11) holds AUX\_EN low while the detector itself is starting (SBVS187G 7.4.2, fig 22), so a hot-plugged AUX-HV can neither enable the LM74800 nor trip the inhibit before the decision is valid.

AUX-HV POWER-GOOD



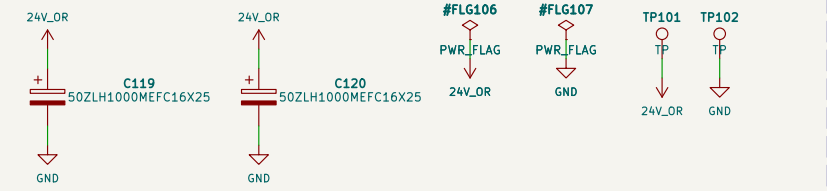
PG\_HVAUX is driven actively high by AUX-HV when good (interfaces.AUX): 100k pull-down here, so a dead or unplugged AUX-HV reads not-good. 10k + BAT54S clamp toward CTRL.

ORED 24 V BUS 24V\_OR



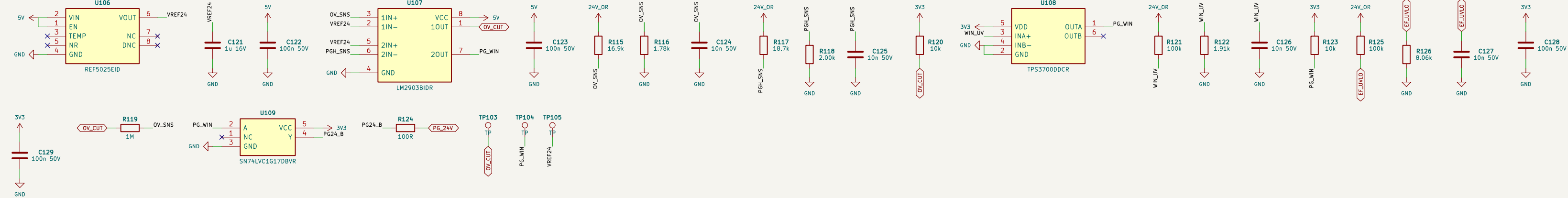
Cabinet feeds share by ideal-diode ORing; AUX-HV joins only through its LM74800 switch.

## SURGE RESERVOIR



2 x 1000 uF 50 V low-impedance electrolytic absorb the surge charge (check surge, every build).  
Design surge IEC 61000-4-5 0.5 kV / 12 ohm / 141.7 A, Q 3.00 m into 2 x 1000 uF (C\_min 1.28 mF, ESR\_max 32 mOhm)  
-> bus and every branch can = 30.07 V [TVS idle below 36.7 V; worst load margin 1.9 V, UCC14243 32 V).  
0.5 kV / 2 ohm line-to-line (EN IEC 61000-6-2 only for 24 V cables > 30 m) would reach 39 V: NOT covered -  
keep the cabinet 24 V wiring < 30 m or fit an external SPD. The OV cut-off may open the eFuses for ~1.3 ms.

24 V WINDOW: OV CUT-OFF, PG\_24V, EFUSE UVLO



D-013: window 21.6–26.0 V; nothing downstream above 26.45 V steady. Worst-case stacks (corners of every trace):  
OV cut-off = LM2930B-1 vs REF502SE: ref 2.5 V +/- 0.066 % (0.025 % + 2.5 ppm/K x 165 K). VIO +4/-mV, IB 50 nA,  
16.9k/1.78k  $\Omega$ , 1 uA ppm/kV x 65 K each. 1 m hysteresis to VOL: 0-0.55 V -> 26.131-26.417 V -> OV CUT -> OV all  
8 efuses (8S=14 uA), PC24W: 18.7% / 200-250 V -> 26.740-26.012 V AND low effuse TPS3D07 INA:  
1.195-1.199 mV, 100k/1.91k 20.5k -21.4 k V, Chn UV (sheet 08) 18.33-19.07 uA efuse UWL (TPS1602 1.176V - 1.224 V  
1.09-1.15 V, 100k/8.00k  $\Omega$ , 1 uA), 15.72-16.48 V on, 14.57-15.48 V off; +24V (CTRL) is the logic input at a hand-over.  
TSV SMCJ33CA (VBR = 36.7 V) stays idle in the design surge - see the surge reservoir.

Not bench-validated. Dual-channel chain gate- and single-fault-checked on every build.  
Sheet 2  
Schematics authored by Chinmoy Bhuyan  
Domains: PELV, CANB, MCAN, RSA, RSB, DI field, PE - only isolators cross (checked)

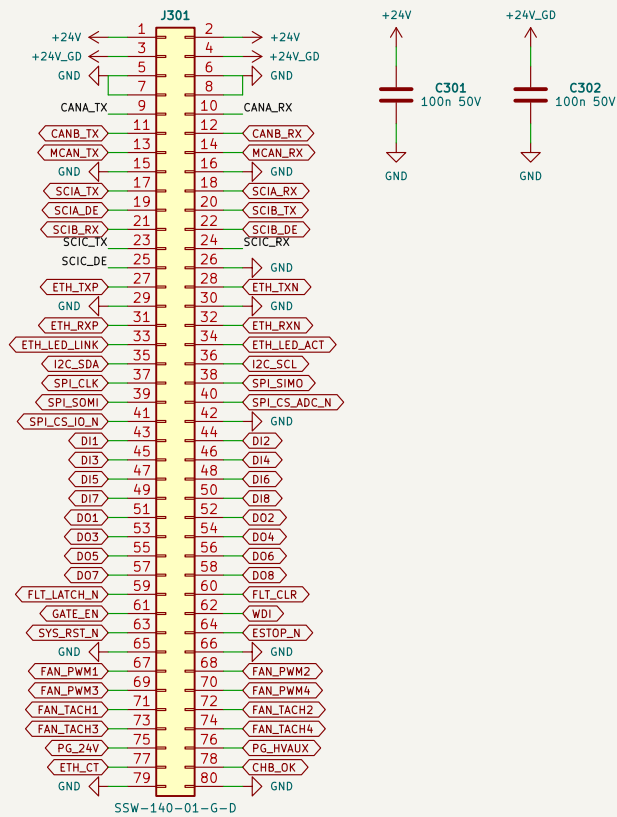
Sheet: /01 24 V entry, protection and window/  
File: 01\_power\_in.kicad\_sch

Title: SYS-10-AUX - 24 V entry, protection and window

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| Size: A2            | Date: 2026-10-04 | Rev: E0  |
| KiCad E.D.A. 10.0.6 |                  | Id: 2/13 |

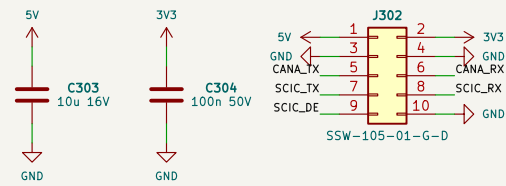


CTRL-C2000 BOARD-TO-BOARD (INTERFACES.SYS)



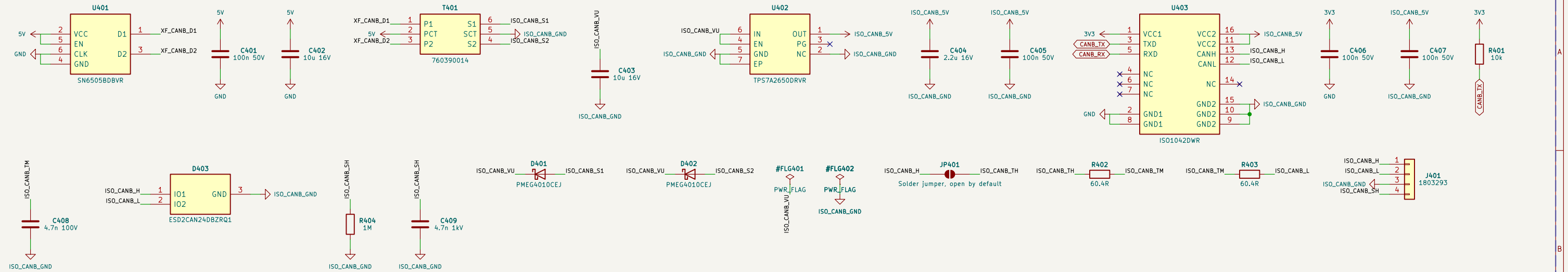
Samtec SSW-140-01-G-D socket mates the TSW-140-07-G-D header on CTRL (both catalogs list the pair; TSW -07 post 5.84 mm inside SSW insertion depth 3.68-6.35 mm). 4.7 A/pin; +24V and +24V\_GD 2 pins each. No I2C pull-ups here (CTRL has them). Pull-ups / pull-downs sit next to each driver on its sheet.

BMU-GW SOCKET (INTERFACES.GW)



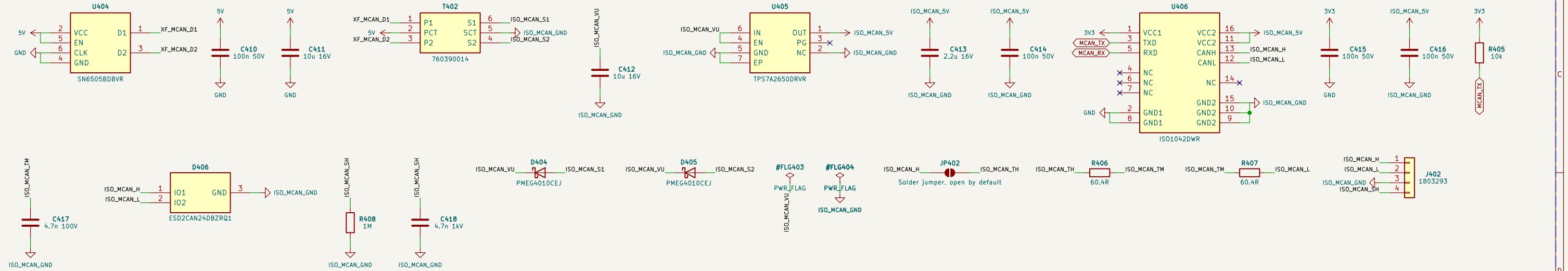
CANA -> CAN\_TX/RX, SCIC -> RS485\_D/R/DE; RS485\_RE\_N tied low here: the SYS contract carries no SCIC\_RE, so the receiver stays on.

### CANB MODULE BUS (ISOLATED)



ISO1042 5 V<sub>RMS</sub>, own isolated 5 V. Split termination 2 x 60.4R + 4.7n, closed by JP.  
Terminal: 1 CANH, 2 CANL, 3 isolated GND, 4 shield. Port supply 760390014 + TPS7A2650: 5.0 V +/-1 %.

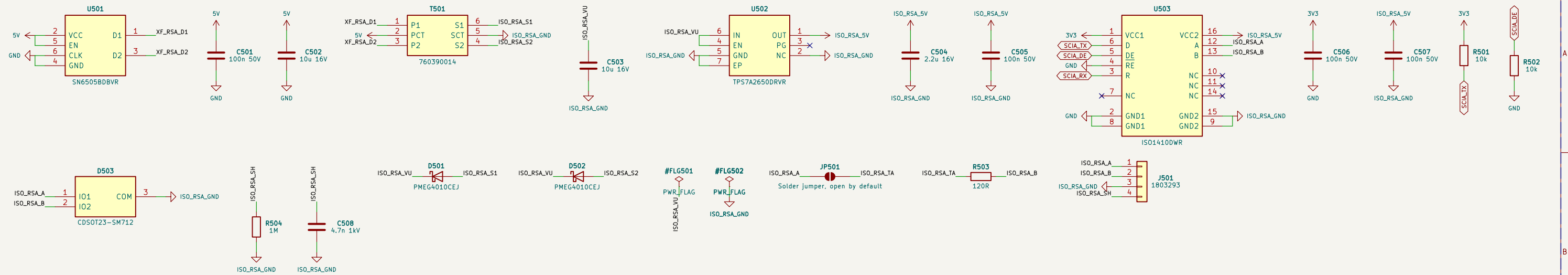
MCAN CAN FD SERVICE PORT (ISOLATED)



ISO1042 5 Vrms, own isolated 5 V. Split termination 2 x 60.4R + 4.7n, closed by JP.  
Terminal: 1 CANH, 2 CANL, 3 isolated GND, 4 shield. Port supply 760390014 + TPS7A2650: 5.0 V +/- 1 %.

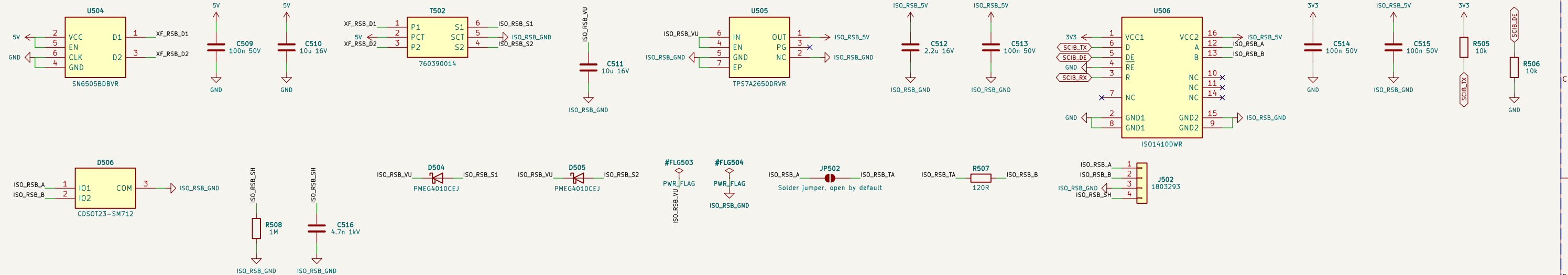
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|-----------------------------------------------------------------------------------------------------------------------------|------------------|----------|
| Not bench-validated. Dual-channel chain gate- and single-fault-checked on every build.<br>Sheet 5                           |                  |          |
| Schematics authored by Chinmoy Bhuyan<br>Domains: PELV, CANB, MCAN, RSA, RSB, DI field, PE - only isolators cross (checked) |                  |          |
| Sheet: /04 Isolated CAN ports (CANB, MCAN)/<br>File: 04_can.kicad_sch                                                       |                  |          |
| <b>Title: SYS-10-AUX - Isolated CAN ports (CANB, MCAN)</b>                                                                  |                  |          |
| Size: A3                                                                                                                    | Date: 2026-10-04 | Rev: E0  |
| KiCad E.D.A. 10.0.6                                                                                                         |                  | Id: 5/13 |

RS-485 PORT 1 = SCIA (ISOLATED)

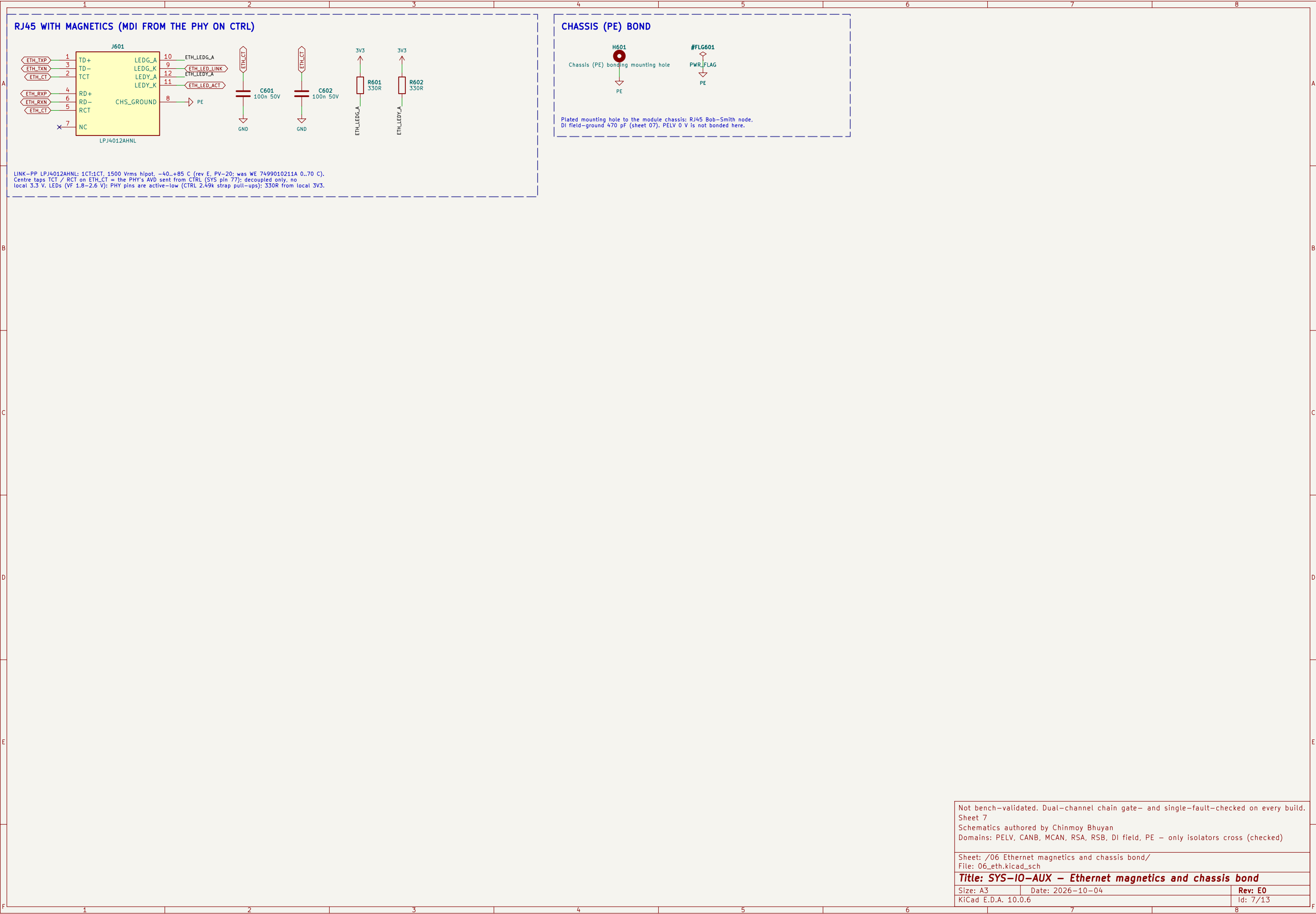


ISO1410 5 kVrms half duplex, own regulated isolated 5.0 V, fail-safe receiver always on (-RE = GND).  
120R termination by JP. Terminal: 1 A, 2 B, 3 isolated GND, 4 shield (1M || 4.7n).

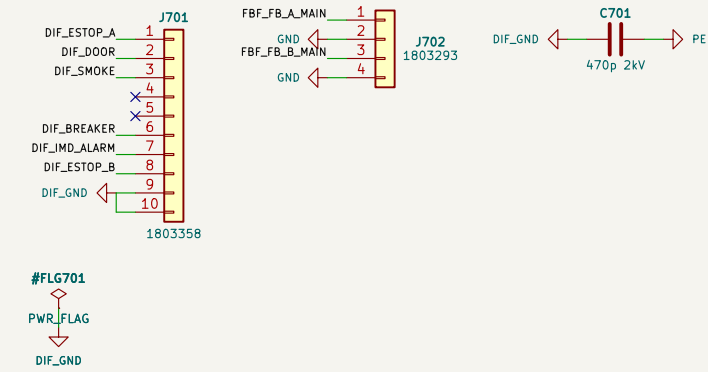
RS-485 PORT 2 = SCIB (ISOLATED)



ISO1410 5 kVrms half duplex, own regulated isolated 5.0 V, fail-safe receiver always on (-RE = GND).  
120R termination by JP. Terminal: 1 A, 2 B, 3 isolated GND, 4 shield (1M || 4.7n).

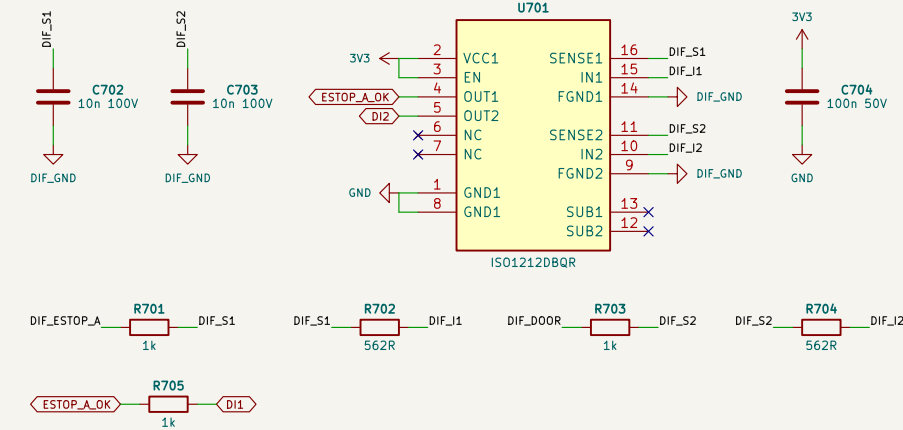


## DI FIELD TERMINALS (INTERFACES.DI)



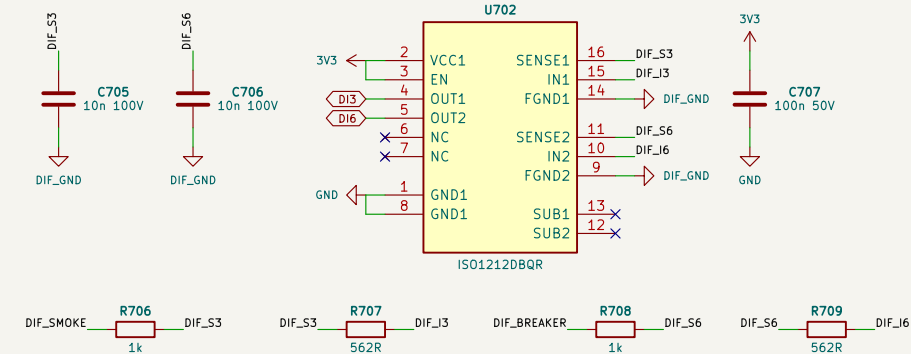
J701 pin n = Din for the six external inputs (4, 5 unused), 9/10 = field 0 V DIF\_GND (470 pF / 2 kV to PE).  
 FB\_A\_MAIN (DI4) / FB\_B\_MAIN (DI5) are module-internal, PELV: J702 1 FB\_A, 2 GND, 3 FB\_B, 4 GND.  
 ISO1212 channels of one package share only functional insulation (+/-60 V, SLLSEY7G 6.1), so DI4 + DI5  
 sit together in U703 with both FGND on GND; U701 / U702 / U704 carry only DIF\_GND channels.

DI1 ESTOP\_A, DI2 DOOR



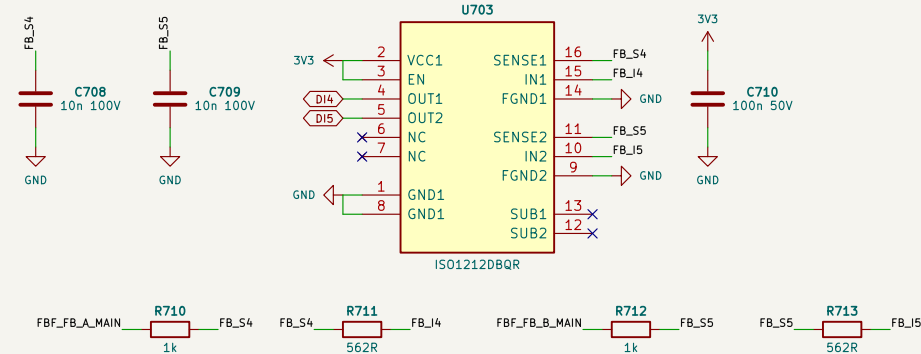
RTHR 1k MELF 0.25 W, RSENSE 562R, CIN 10n: VIH <= 10.95 V, VIL >= 8.7 V, 2.25 mA -> type 1 and 3.  
SUB pins go to a floating 2 x 2 mm copper island only (datasheet), never to FGND. FGND = DIF\_GND.  
DI1 = E-stop channel A: drives the chain directly; DI1 to CTRL through 1k.

DI3 SMOKE, DI6 BREAKER



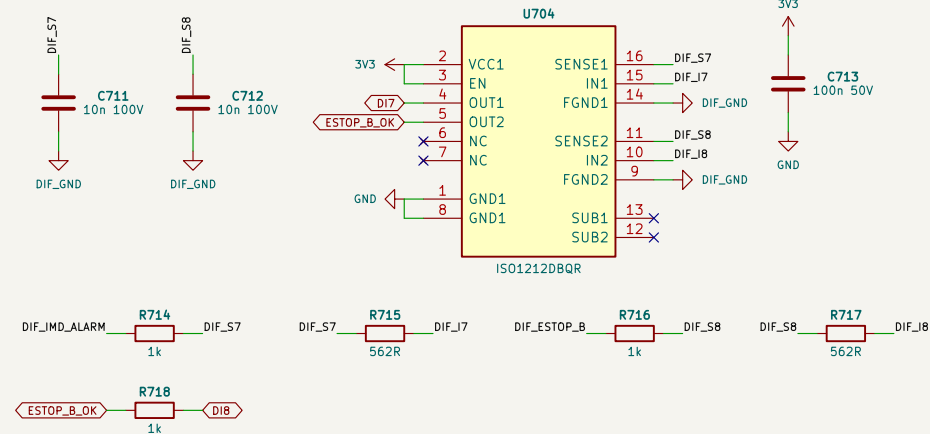
RTHR 1k MELF 0.25 W, RSENSE 562R, CIN 10n: VIH <= 10.95 V, VIL >= 8.7 V, 2.25 mA -> type 1 and 3.  
SUB pins go to a floating 2 x 2 mm copper island only (datasheet), never to FGND. FGND = DIF\_GND.

DI4 FB\_A\_MAIN, DI5 FB\_B\_MAIN



RTHR 1k MELF 0.25 W, RSENSE 562R, CIN 10n: VIH  $\leq 10.95$  V, VIL  $\geq 8.7$  V, 2.25 mA  $\rightarrow$  type 1 and 3.  
SUB pins go to a floating 2 x 2 mm copper island only (datasheet), never to FGND. FGND = GND.

DI7 IMD\_ALARM, DI8 ESTOP\_B



RTHR 1k MELF 0.25 W, RSENSE 562R, CIN 10n: VIH <= 10.95 V, VIL >= 8.7 V, 2.25 mA -> type 1 and 3.  
SUB pins go to a floating 2 x 2 mm copper island only (datasheet), never to FGND. FGND = DIF\_GND.  
D18 = E-stop channel B: drives the chain directly; D18 to CTRL through 1k.

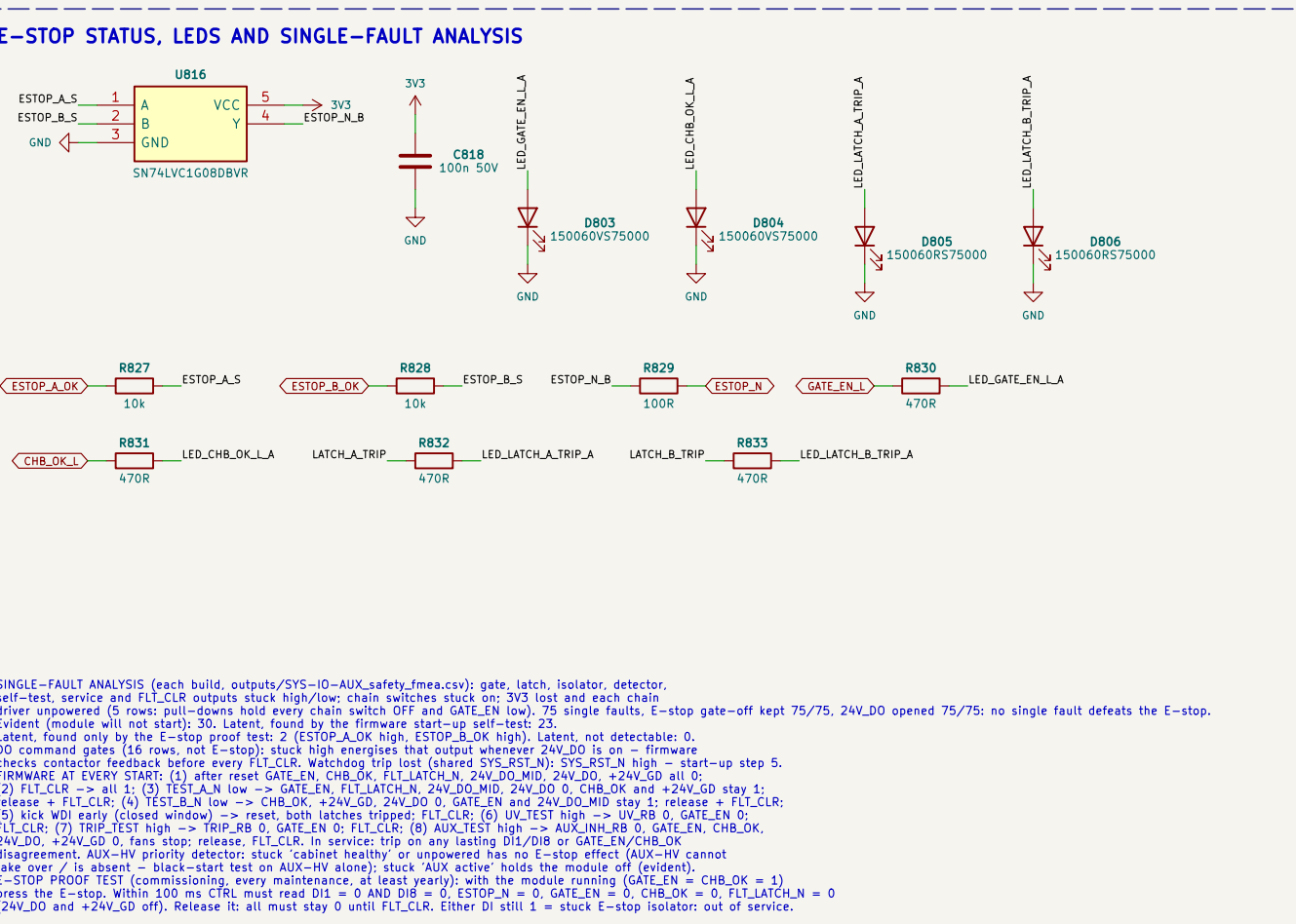
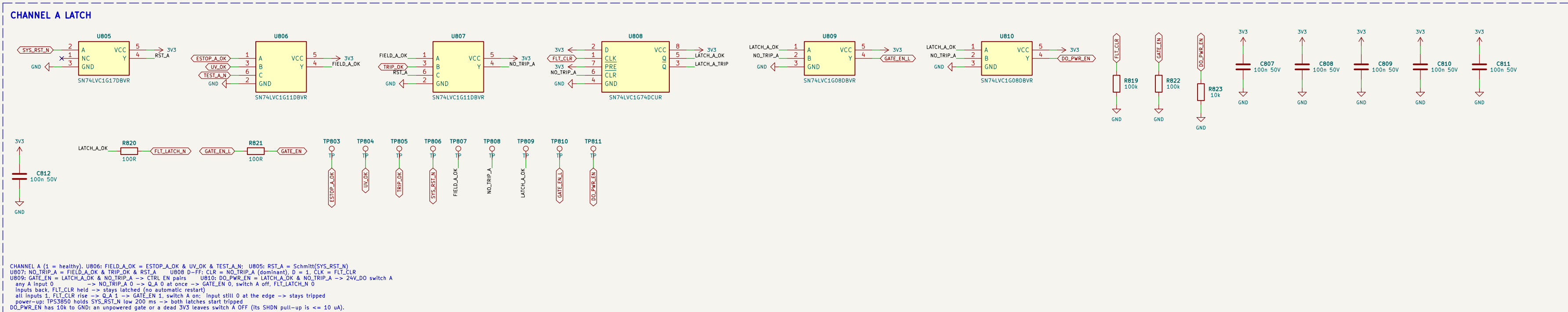
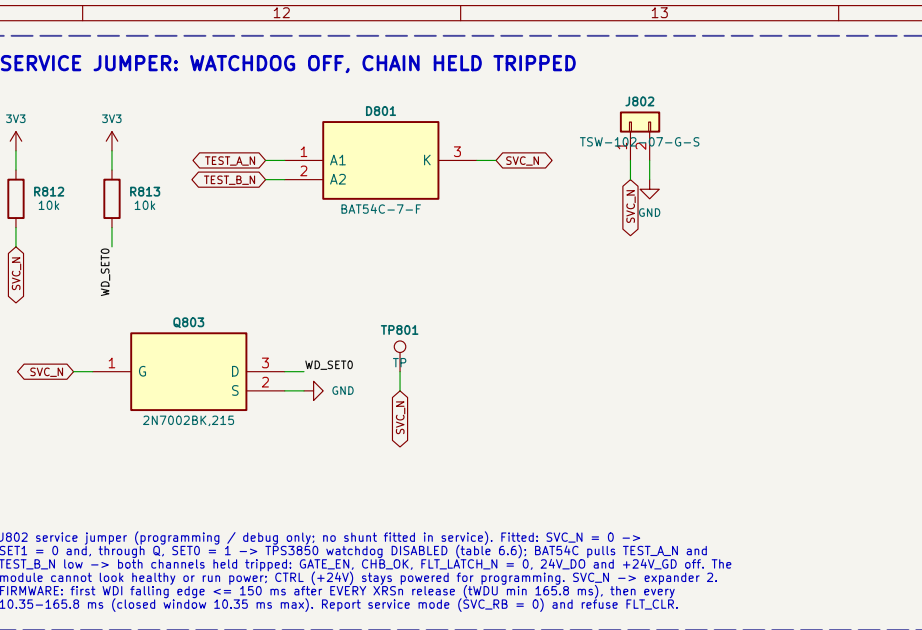
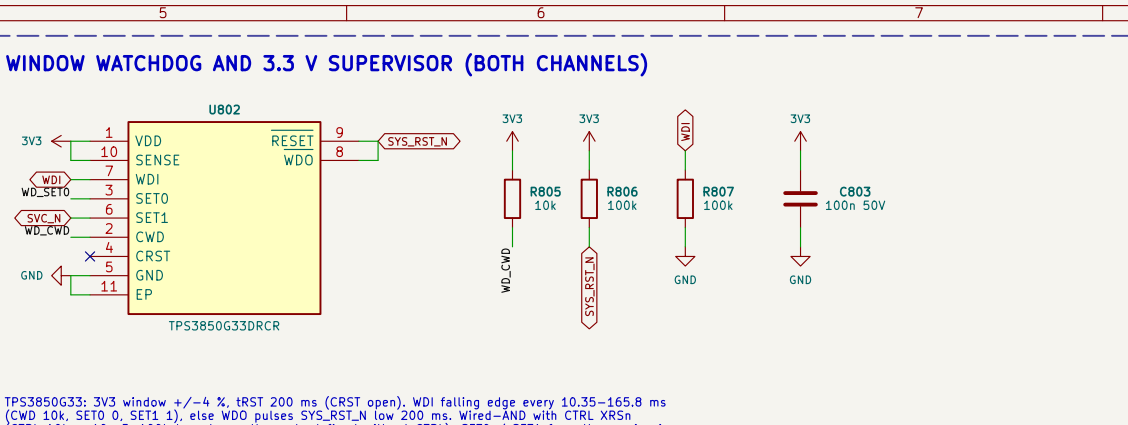
Not bench-validated. Dual-channel chain gate- and single-fault-checked on every build.  
Sheet 8  
Schematics authored by Chinmoy Bhuyan  
Domains: PELV, CANB, MCAN, RSA, RSB, DI field, PE – only isolators cross (checked)

Sheet: /07 Isolated 24 V digital inputs/  
File: 07\_di.kicad\_sch

**Title: SYS-10-AUX – Isolated 24 V digital inputs**

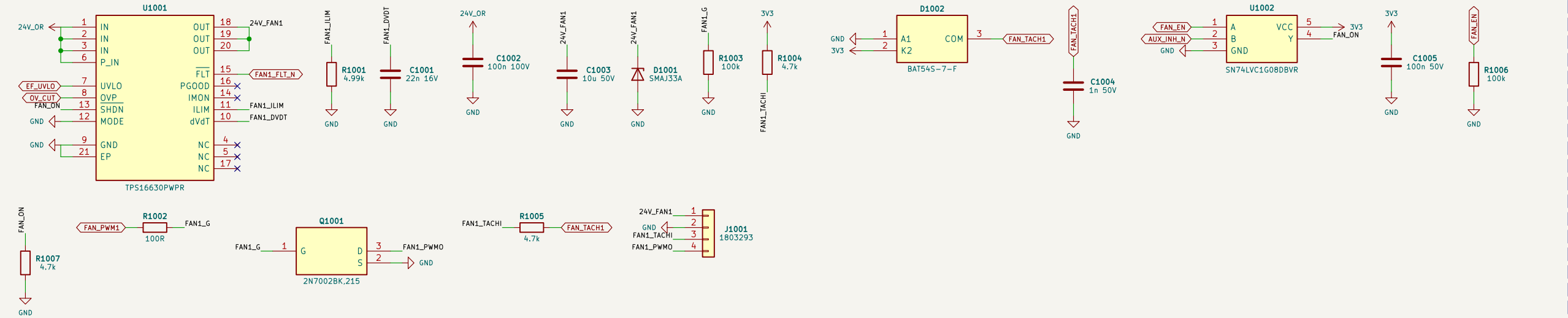
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| KiCad E.D.A. 10.0.6 |                  | Id: 8/13 |



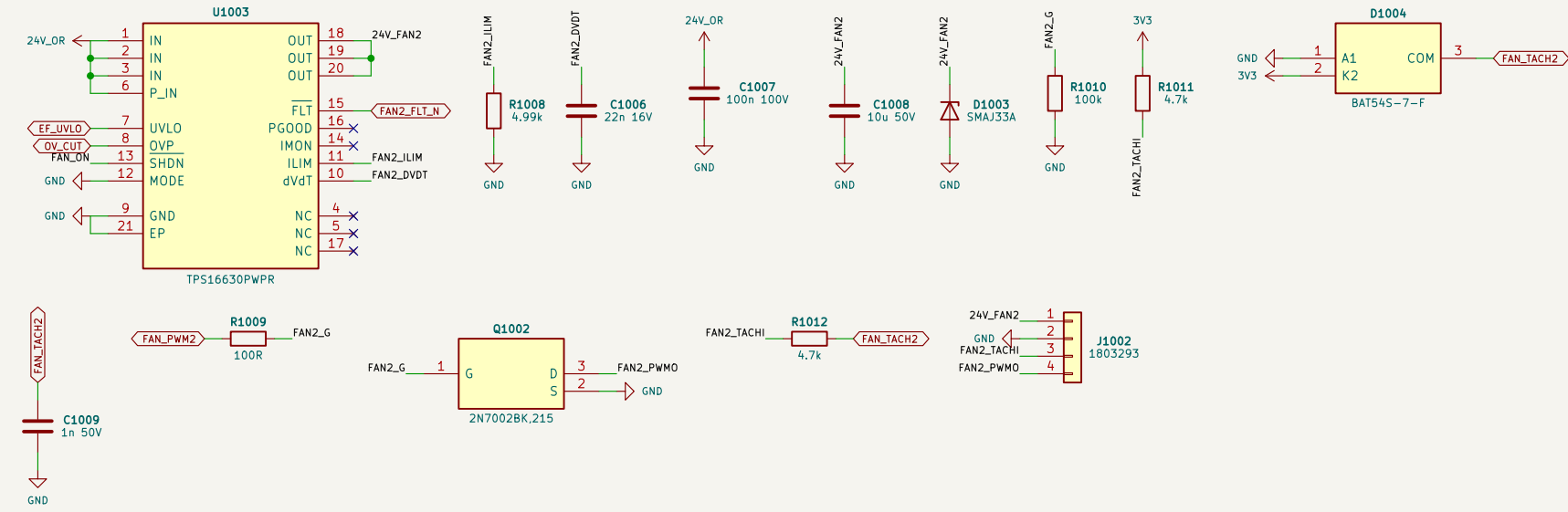


FAN 1

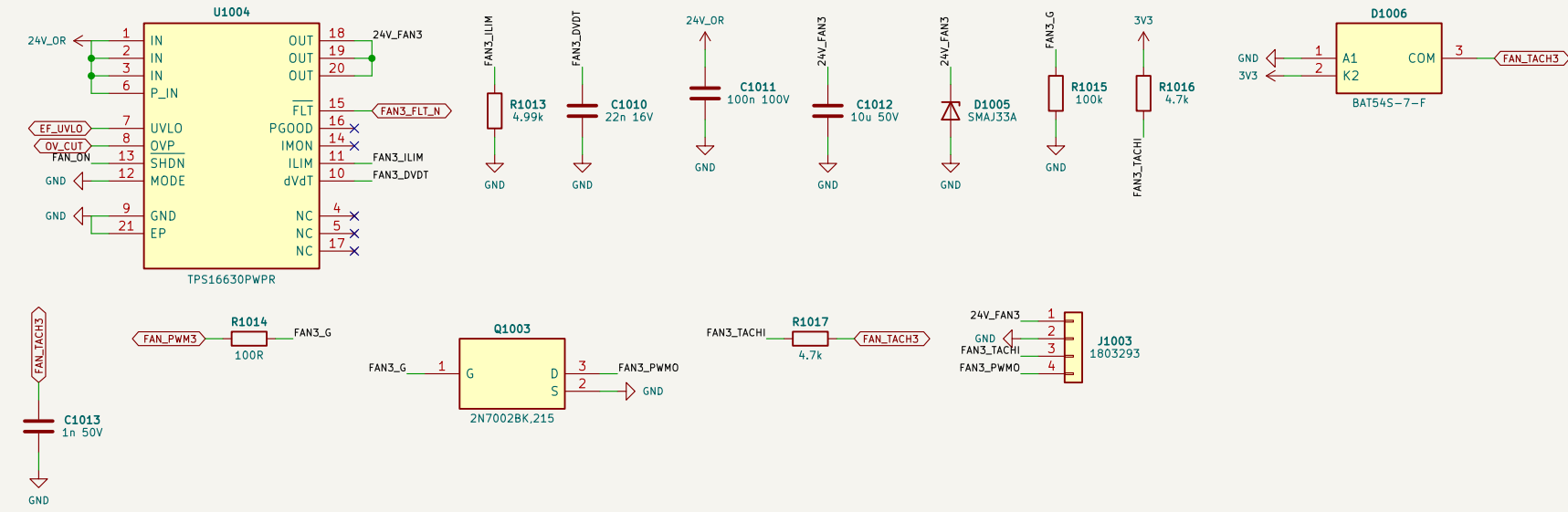


Terminal: 1 +24 V, 2 0 V, 3 tach (blue), 4 PWM (yellow). ILIM 4.99k -> 3.6 A.  
FAN\_ON = FAN\_EN (expander 1 GPB4, 100k pull-down) AND AUX\_INH\_N: fans are OFF until  
firmware sets FAN\_EN and whenever AUX-HV is the source (no firmware in that path);  
4.7k on FAN\_ON keeps all four off if the gate is unpowered.  
PWM inverted: FAN\_PWM high = fan input low; once powered, PWM open = full speed.  
Tach: open collector, 4.7k to 3V3 (0.7 mA < 5 mA), 4.7k + BAT54S clamp.

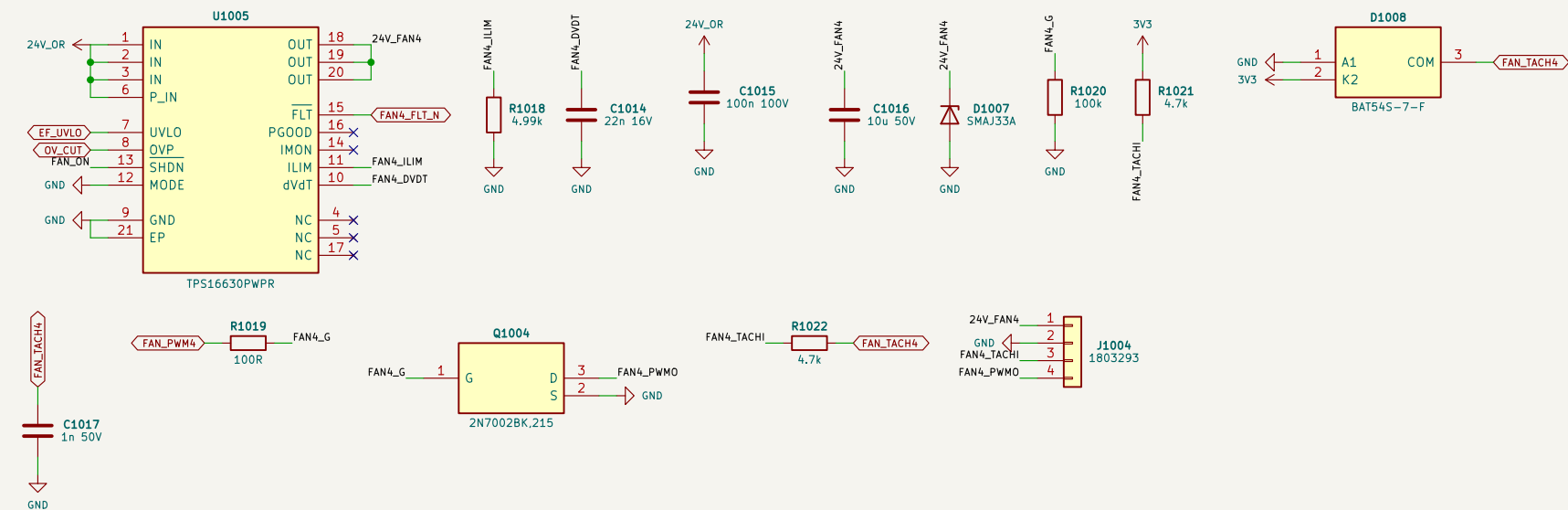
FAN 2



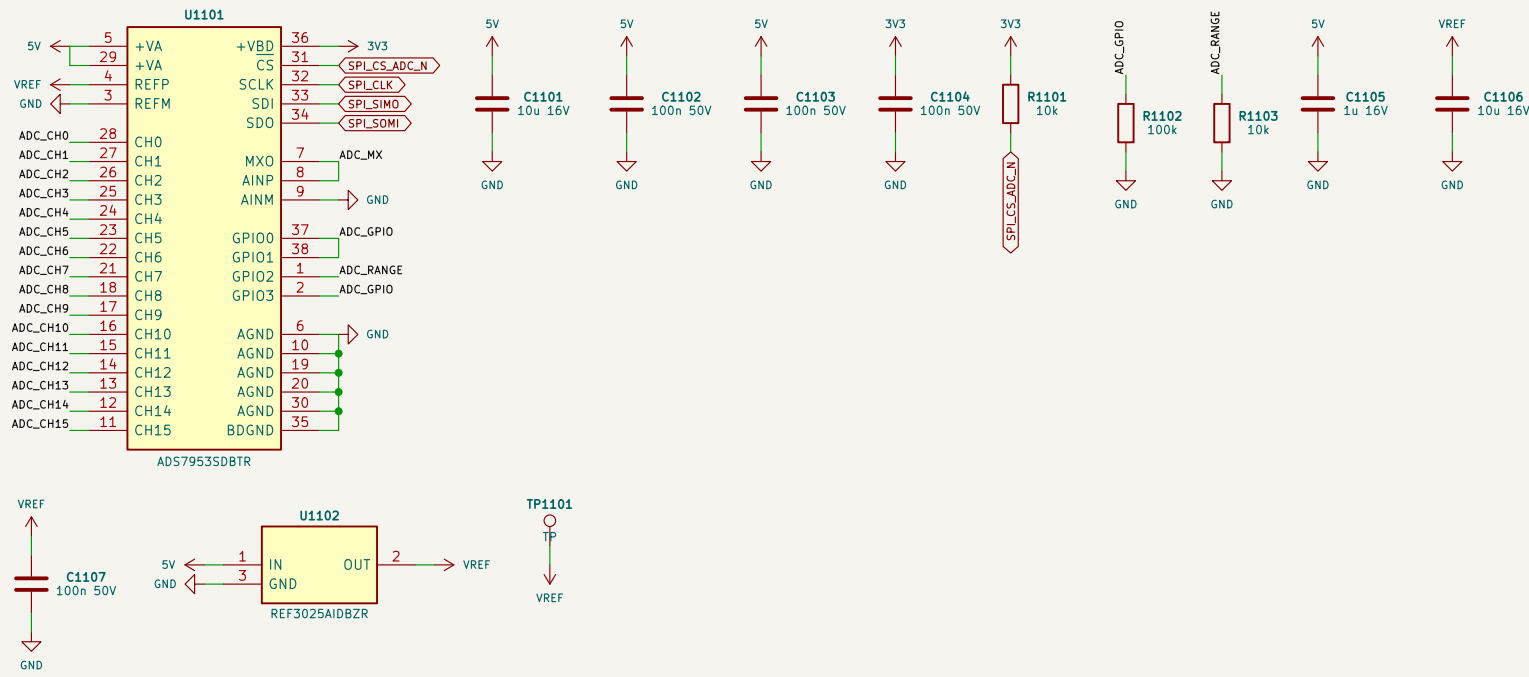
FAN 3



FAN 4

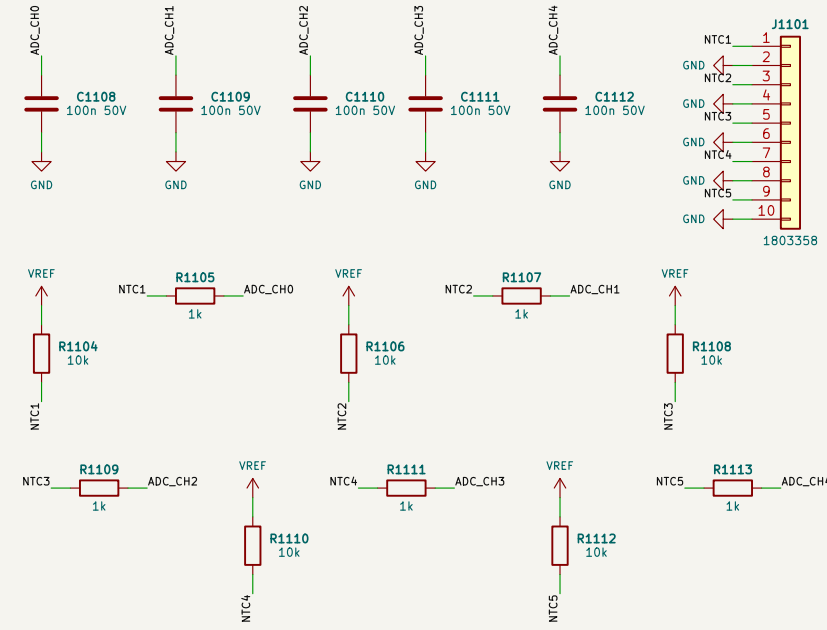


ADC AND REFERENCE



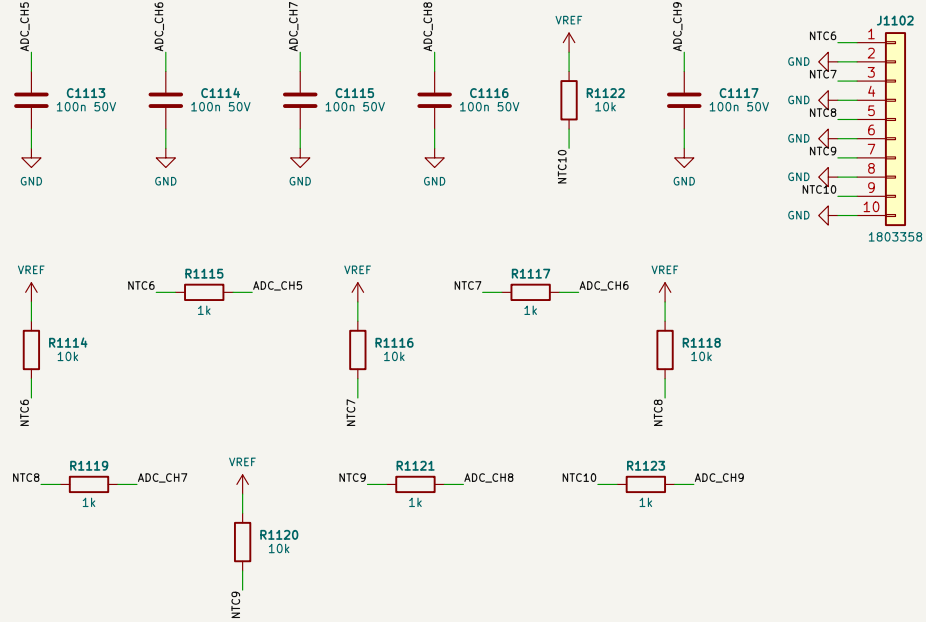
ADS7953: +VA 5 V, +VBD 3.3 V, range 1 (0..VREF, GPIO2 pulled low), MX0 -> AINP.  
GPIO0/1/3 inputs by default, pulled low. SDO tri-states when CS is high.

NTC1-NTC5



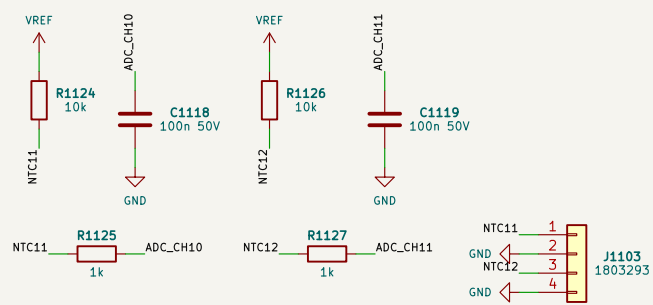
NTC: B57703M0103 (10k, B 3988), VREF 2.5 V -> 10k 0.1 % -> NTC: 1k + 100n -> ADC (ratiometric).  
Codes (12 bit): -40 C 3978, 25 C 2048, 125 C 135, 150 C 74, Open > 4040, short < 40:  
both are outside the -40..150 C band, so a broken or shorted sensor is distinguishable.

NTC6-NTC10



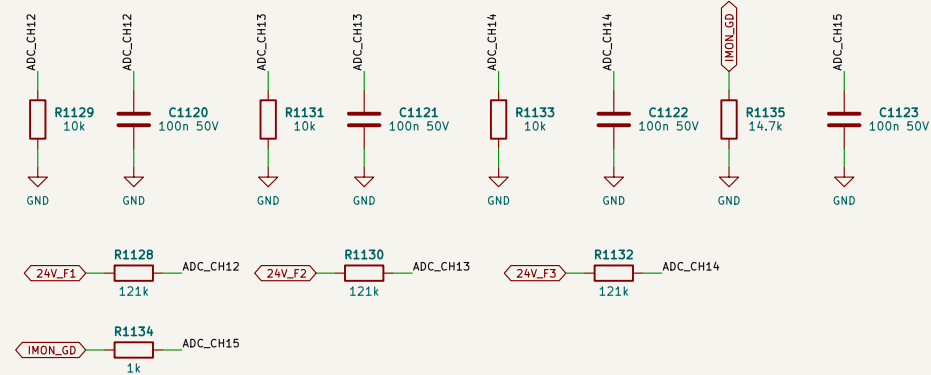
Terminal pairs: odd = NTC, even = 0 V return.

NTC11-NTC12



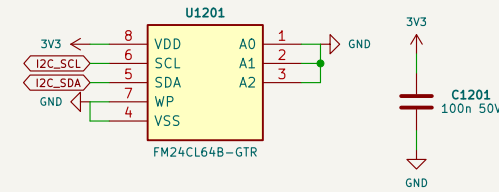
Terminal pairs: odd = NTC, even = 0 V return.

SUPPLY MONITORS



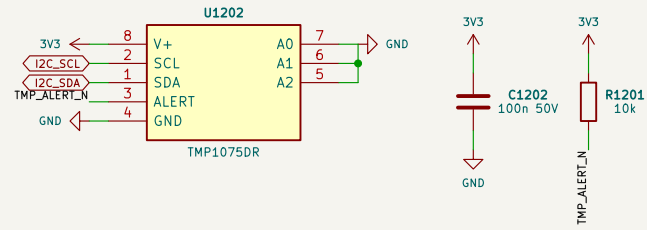
Feeds: 121k / 10k -> full scale 32.7 V (CH12 feed 1, CH13 feed 2, CH14 AUX-HV).  
CH15: IMON\_GD 27.9 uA/A x 14.7k -> 6 A = 2.46 V (gate-drive bias of all power boards).

SERVICE F-RAM



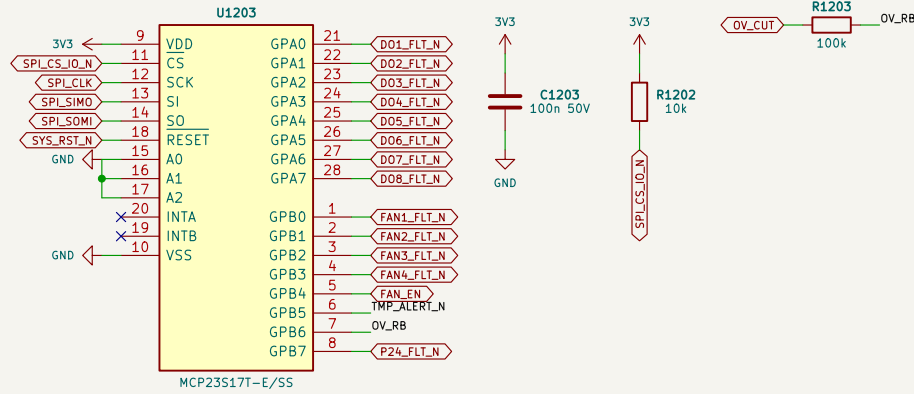
64 kbIt F-RAM 1e14 write cycles; an hours counter written every minute for 20 years is -1e7 writes. WP low (writable). I2C pull-ups and the 0x57 EEPROM are on CTRL.

BOARD TEMPERATURE



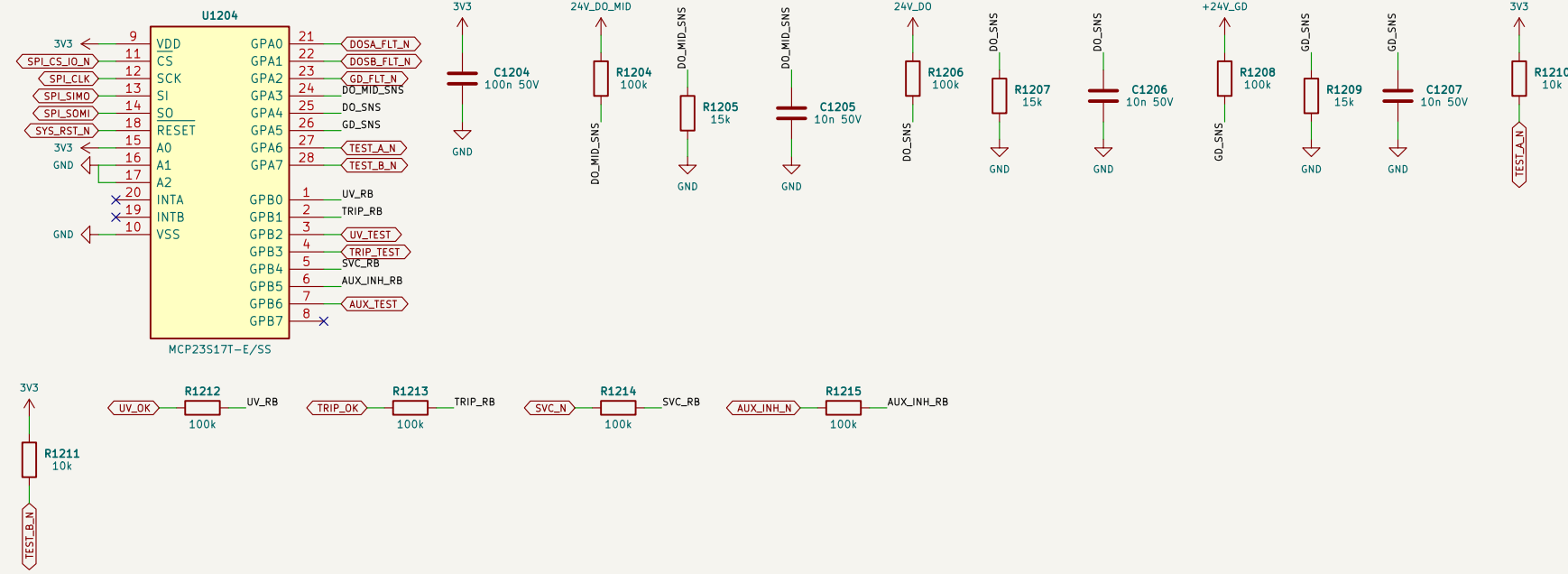
TMP1075 at 0x48 (no clash with 0x50 / 0x57): ALERT -> expander 1 GPB5.

EXPANDER 1 (ADDRESS 0): OUTPUT-DRIVER AND FAN FAULTS



GP0-7 = DO1-8 FLT (TPS272C45D). GP0-3 = fan eFuse FLT. GP4 = FANLEN out. GP5 = TMP ALERT. GP6 = OV\_CUT readback. GP7 = +24V eFuse FLT. Inputs: enable GPPU pull-ups. Reset = SYS\_RST\_N. MCP23S17 at 3.3 V is rated -40..+85 C (DS20001952C p1: +125 C only at 4.5-5.5 V): enough for PV-20.

EXPANDER 2 (ADDRESS 1): CHAIN SELF-TEST AND SWITCH READBACK



GP0-2 = DOS A / DOS B / GD eFuse FLT. GP3-5 = 24V\_DO\_MID / 24V\_DO / +24V\_GD present (100k/15k: 21.6 V -> 2.82 V: surge 3.95 V clamp < 0.1 mA). GP6/7 = TEST\_A\_N / TEST\_B\_N outputs (10k pull-ups: idle healthy: driving low can only trip a channel). GP80/1 = UV\_OK / TRIP\_OK readback via 100k. GP82/3 = UV\_TEST / TRIP\_TEST outputs (100k pull-downs: idle off: high forces that detector input into its trip state). GP84 = SVC\_RB (service jumper: fitted = 0). GP85 = AUX\_INH\_RB (0 = AUX-HV is the source, power outputs inhibited). GP86 = AUX\_TEST output (100k pull-down: high forces the inhibit). GP87 spare (enable GPPU).

Not bench-validated. Dual-channel chain gate- and single-fault-checked on every build.  
Sheet 13  
Schematics authored by Chinmoy Bhuyan  
Domains: PELV, CANB, MCAN, RSA, RSB, DI field, PE - only isolators cross (checked)